DonProof HS

Pre-applied, fully bonded HDPE Waterproof Membrane (Sand Finish)

DESCRIPTION

DonProof HS is a high-performance, pre-applied, fully bonded HDPE waterproofing membrane engineered with a multi-layer composite structure to ensure superior waterproofing protection.

The membrane comprises a durable HDPE sheet, a pressure-sensitive adhesive layer, and a unique particulate surface/sand finish layer. DonProof HS are supplied with an integrated selvedge along one edge to facilitate self-adhered overlaps and ensure continuity between adjacent sheets.

DonProof HS is designed for installation on uniform concrete blinding or compacted sand-stone cushion bases, making it ideal for below-ground waterproofing applications, including confined areas and blind-side conditions where direct access is limited.

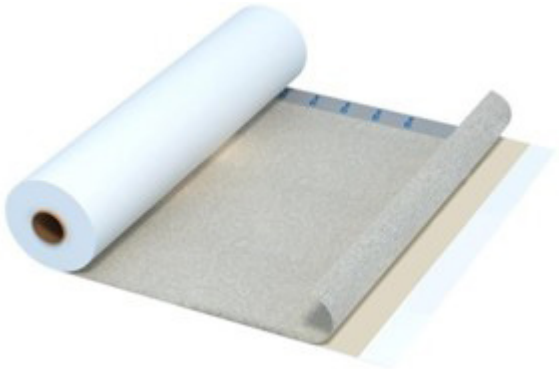
Upon concrete placement, DonProof HS develops a hybrid bonding system with the poured concrete, combining both chemical interaction and physical adhesion. The adhesive layer and sand-finish surface establish intermolecular attraction with the concrete matrix through Van der Waals forces, contributing to chemical-level interlocking, while the granular sand layer promotes mechanical interlocking with the curing concrete to form a continuous and integral interface.

This combined mechanism ensures a robust, fully bonded system that minimizes water migration, prevents lateral water tracking, and delivers reliable, long-term watertight performance.

APPLICATIONS

DonProof HS membrane can be used in:

- Below-grade waterproofing.
- Foundations and basements.
- Protection against underground gas leakage (coal gas, methane, etc.).
- Tunneling (MRT, subways, metro).



TECHNICAL PROPERTIES:

Material:	HDPE + pressure-sensitive adhesive + sand layer
Thickness:	1.2, 1.5 or 2.0 mm
Tensile strength (MD): ASTM D 412, modified	≥ 22 MPa
Elongation: ASTM D 412, modified	≥ 500%
Puncture Resistance: ASTM E 154	1000 N
Peel adhesion to concrete: ASTM D903, modified	1.0 N/mm
Peel resistance: BS EN 1372:2015	3.0 N/mm
Hydrostatic pressure resistance: ASTM D 5385: 1993, modified	≥ 0.7 MPa, (~7 bar) 1 h No water leakage
Water vapor transmission: ASTM E 96	0.4 ng/(m²·s·Pa)
Static load resistance: EN 12730 Method B	No leak at 20 kg (Pass)
Tensile strength (MD): EN ISO 12311-2 Method B	≥ 15 MPa
Elongation: (MD / CMD) EN 12311-2 Method B	≥ 400%

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ADVANTAGES

- Fully bonded system.
- Mechanical adhesion through granular sand finish.
- Provides a water and moisture barrier.
- Excellent mechanical properties.
- Easy and quick application, no priming or fillets required.
- Temporarily UV resistant during application and steel works.
- Trafficable immediately after installation and ready for immediate placing of reinforcing steel, eliminating the need for additional protective layers.
- Displacement resistant, maintains integrity and adhesion even in ground settlement conditions.
- Water-resistant, performs reliably in wet or damp environments without degradation.
- Watertight laps and detailing.
- Hybrid bonding system with the poured concrete, combining both chemical interaction and physical adhesion.

STANDARDS

DonProof HS complies with DIN EN 13967, Type T waterproofing membrane as per the technical properties table.

METHOD OF USE

DonProof HS must be installed by experienced and qualified personnel.

SURFACE PREPARATION

Prior to placement, the blinding surface should be thoroughly cleaned to ensure it is free from any loose materials, sharp projections, or debris that could damage the membrane. All substrates must be smooth, continuous, and free from gaps or voids exceeding 12 mm.

The blinding layer must present a smooth and level surface, free of loose aggregate, sharp protrusions, or irregularities which could damage the membrane. Curved or round surfaces should be avoided.

TECHNICAL PROPERTIES:	
Tear resistance (MD): EN 12310-1 (Nail shank)	≥ 200 N
Tear resistance (CMD): EN 12310-1 (Nail shank)	≥ 500 N
Water tightness to liquid water: EN 1928, Method A	No water penetration through to the upper filter paper (Pass)
Water tightness: DIN EN 1928:2000 (Method B)	Watertight, 7 bar, 72 hr No water has been observed on the unexposed face of membrane
UV resistance: ASTM G154-16	Pass No crazing, chalking, cracking blistering, or color change after 56 days exposure
Resistance to impact: EN 12691	≥ 1500 mm
Lap peel adhesion: ASTM D 1876	1.5 N/mm
Low temperature flexibility: ASTM D 1970	-25 (Pass)
Joint shear resistance: EN 12317-2	≥ 600 N/50 mm

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APPLICATION

For horizontal surfaces, DonProof HS membrane should be unrolled with the white HDPE film side facing the substrate. To avoid excessive membrane buildup, end laps should be staggered.

For vertical surfaces, DonProof HS membrane should be applied with the white HDPE film side facing the substrate (flush pointed Block work, plum concrete wall or plywood shuttering). DonProof HS should be mechanically fastened vertically using flat-headed fixings that are appropriate for the specific substrate. The membrane can be installed in convenient lengths, depending on site conditions. The top edge of the membrane should be secured using a batten or suitable fixing approximately 50 mm below the top edge. To ensure the membrane lies flat against the surface, fixings should be placed at intervals of approximately 600 mm centers. These fixings may be made through the selvedge area, allowing subsequent sheets to overlap and be firmly rolled, ensuring the fasteners are covered by the next membrane layer.

For horizontal and vertical surfaces, the plastic release liner over the selvedge area must be left in place until the overlapping procedure is completed.

Succeeding membrane sheets should be accurately positioned to provide a 75 mm overlap along the self-adhesive selvedge.

Before bonding, it must be ensured that the underside of the succeeding sheet and the surface of the release liner are clean, dry, and free from any contamination, wiping with a damp cloth if necessary.

The plastic release liner is then peeled back and progressively removed as the two membrane layers are bonded together. All overlaps should be thoroughly pressed using a steel roller to eliminate gaps or creases, enabling a consistent, watertight seal, a continuous, crease-free bond must be ensured.

For areas without selvedge, laps should be bonded using a suitable double-sided adhesive tape (Donproof Tape H200), and the bonded joints must be rolled firmly to secure the system.

DAMAGE REPAIR

After the placement of the bottom reinforcement bars, the membrane must be inspected for any damage or punctures. Any identified damage should be repaired immediately. Damaged areas must be patched with an oversized piece of membrane, which is to be applied over a clean and dry surface.

For minor damage, such as surface scratches, repairs may be carried out directly by applying sanded tape (Donproof Tape H100), centrally over the affected area.

REMARKS

- During damp or humid conditions—such as after water exposure—the selvedge area may be gently warmed using a hot air gun or similar device. This process helps remove moisture or condensation and enhances the initial adhesion of the lap joint.
- For installation of DonProof HS, a dry surface is not required; however, all standing or ponded water must be removed prior to membrane application.
- DonProof HS is suitable for installation at temperatures of -5°C and above. In cold weather conditions, the bonding edge must be pre-heated using a hot air gun.

PACKAGING

Thickness: 1.2, 1.5 mm or 2.0 mm
Width: 1 m + 1%
Length: 20 m

STORAGE

DonProof HS rolls must be stored at temperatures between 5°C and 45°C, away from direct sunlight in cool and dry conditions.

DonProof HS must be stored vertically as a single layer. Avoid contact with acids,alkalis,oils,organic solvents,etc.

If these conditions are exceeded, contact DCP Technical Department for advice.

CAUTIONS

HEALTH AND SAFETY

DonProof HS should be welded in a well-ventilated area.

For further information, refer to the Material Safety Data Sheet.

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- Sealants.
- Waterproofing.
- Adhesives.
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- Building products.
- Structural strengthening.

Note:
We endeavour to ensure that any information, advice or recommendation we may give in product literature is accurate and correct. However, because we have no control over where and how products are applied, we cannot accept any liability arising from the use of the products.